



**JABATAN PEMBANGUNAN KEMAHIRAN
KEMENTERIAN SUMBER MANUSIA
ARAS 7 & 8 BLOK D4, KOMPLEKS D
PUSAT PENTADBIRAN KERAJAAN PERSEKUTUAN
62530 PUTRAJAYA**

KERTAS PENERANGAN (*INFORMATION SHEET*)

KOD DAN NAMA PROGRAM / <i>PROGRAM CODE & NAME</i>	Z-009-1: 2015CORE ABILITIES	
TAHAP / LEVEL	4 (FOUR)	
NO. DAN TAJUK KEBOLEHAN TERAS / <i>CORE ABILITY NO. AND TITLE</i>	CA02 HEALTH, SAFETY AND ENVIRONMENTAL AWARENESS	
NO. DAN PENYATAAN KEBOLEHAN / <i>ABILITY NO. AND STATEMENT</i>	02.01 ENSURE HEALTH CONCIOUSNESS 02.02 ENSURE SAFETY CONCIOUSNESS 02.03 ENSURE ENVIRONMENTAL AWARENESS PROGRAM	
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TAJUK/TITLE: ENVIRONMENTAL AWARENESS PROGRAM

TUJUAN/PURPOSE:

This information sheet is intended to describe the implementation of environmental awareness programs that can be conducted at workplace for workers and also public. The person who is competent in this CA shall be able to ensure environment awareness programme at workplace and public.

PENERANGAN/INFORMATION:

Environmental awareness program objectives

Environment is degrading at a much faster pace than our imagination. Most of this mess is

caused by human activities. The damage is both at global and regional level. The rapid development of economic activities, especially in the agriculture and industrial sectors, have accelerated the process of urbanisation and infrastructure development. The changes of ecosystem started with changes in the forest ecosystem which then carried on to agricultural areas. Later, these agricultural areas changed into urban areas. In some ways, rapid changes are setback because the ecosystem is disturbed. This leads to many environmental problems such as flash flood, coastal erosion, water supply disruption, loss of biodiversity, loss of breeding areas for marine life and increasing temperature.



Therefore whatsoever wrong has been done must be rectified. Maintenance of ecological balance and a pristine environment has been of utmost importance. A regime of strict protocols for employees and contractors to lessen or eliminate the impact of emissions or waste on the environments in which they work need to be devised. To protect and manage environment, it is

imperative to have a sound environmental education. It is a way to teach people and societies on how to use the present and future resources optimally.

Environmental education is a learning process that increases knowledge and awareness about the environment and associated challenges, develops the necessary skills and expertise to address the challenges, fosters attitudes, motivations and commitments to make informed decisions and take responsible action. The components of environmental education are:

- awareness and sensitivity to the environment and environmental challenges;
- knowledge and understanding of the environment and environmental challenges;
- attitudes of concern for the environment and motivation to improve or maintain environmental quality;
- skills to identify and help resolve environmental challenges.

Environmental awareness needs to be created through formal and informal education to all sections of the society. Everyone needs to understand it because, environment belongs to all and, every individual matters when it comes to conservation and protection of environment. Various stages and methods that can be useful for raising environmental awareness in different section:

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i) Among students through education

Environmental education must be imparted to the students right from the childhood stage. It is a welcome step that now all over the country we are introducing environmental studies as a subject at all stages including school and college level.

ii) Among the Masses through mass-media

Media can play an important role to educate the masses on environmental issues through articles, environmental rallies, plantation campaigns, street plays, real eco-disaster stories and success stories of conservation efforts. TV serials can be effective in propagating the seeds of environmental awareness amongst the viewers of all age groups.

iii) Among the industries, employers and workers

Since they are the main contributors to environmental effects to the world, awareness and education will help them in understanding the importance and the right way to conserve the environment. Campaign and training is one of the methods in doing this.

iv) Among the planners, decision-makers and leaders

Since this elite section of the society plays the most important role in shaping the future of the society, it is very important to give them the necessary orientation and training through specially organized workshops and training programmes.

Publication of environment, related resource material in the form of pamphlets or booklets published by Department of Environment can also help in keeping this section abreast of the latest developments in the field.

Environmental awareness proves important for several reasons; it fosters a sense of connection to the natural world, promotes sustainable development and encourages conservation of irreplaceable natural resources and vulnerable plant and animal species. Environmental awareness essentially serves as an educational tool, helping the industries especially the management to understand the importance of controlling by means of reducing or eliminating the harmful impacts of work activities by the workers to environment. The need to spread environmental awareness is enormous in the context of successfully addressing environmental problems.

As for the industries, worker's responsible for environmental management should be properly qualified and have the awareness, knowledge and skills to implement

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environmental best practices in accordance with regional and international standards. Various training programs may be appropriate for them to undergo before they embark upon staff training, such as ISO 14001 and EMS foundation courses, or other.

As referring to the Environmental Management System ISO 14001:2004, it is the responsibility of the concerned manager to identify needs, develop and implement comprehensive awareness program. Environmental Management System (EMS) coordinator shall review the program every quarter. The development of a culture of continuous environmental education and training will provide greater awareness capability and motivation to employees and enable them to conduct their daily operational activities in an environmentally responsible manner. Such a program can be a strong motivational tool and can bind workers closer to the company's environmental ethic.

The Environmental Awareness programs allows all workers to identify factors associated with the risks to the environment. It enables an organisation to establish an environmental management system allowing workers to implement practical steps to protect the environment.

Awareness programs (Industries Level)

What employers and workers should know about environment during their everyday work and how their action and prevention can contribute in conserving the environment? Main agenda that they need to be taught and aware of is the environmental pollutions.

Global Warming



Global warming is the term used to describe a gradual increase in the average temperature of the Earth's atmosphere and its oceans, a change that is believed to be permanently changing the Earth's climate. The increased volumes of carbon dioxide and other greenhouse gases released by the burning of fossil fuels, land clearing, agriculture, and other human activities, are believed to be the primary sources of the global warming that has occurred over the past 50 years.

In some manufacturing industries, greenhouse gases like hydrofluorocarbons, methane and nitrous oxide released as by-products of certain manufacturing industry processes, which are adversely affect the ozone layer, and leading to global warming.

Air pollutions



Generally any substance that people introduce into the atmosphere that has damaging effects on living things and the environment is considered air pollution. Manufacturing industries release large amount of carbon monoxide, hydrocarbons, organic compounds, and chemicals into the air thereby depleting the quality of air.

Ammonia is a very common by-product from agriculture related activities and is one of the most hazardous gases in the atmosphere. Use of insecticides, pesticides and fertilizers in agricultural activities has grown quite a lot. They emit harmful chemicals into the air.

Mining is a process wherein minerals below the earth are extracted using large equipments. During the process dust and chemicals are released in the air and causing massive air pollution. This is one of the reasons, which is responsible for the deteriorating health conditions of workers and nearby residents.

Water pollutions



Water pollution is the contamination of natural water bodies by chemical, physical, radioactive or pathogenic microbial substances. Chemical water pollutants are generally atoms or molecules, which have been discharged into natural water bodies, usually by activities of humans. Common examples of such chemical water pollutants are mercury emanating from

mining activity, certain nitrogen compounds used in agriculture, chlorinated organic molecules arising from sewage or water treatment plants or various acids which are the externalities of various manufacturing activities. Physical water pollutants are either (a) much larger particles or (b) physical factors such as temperature change, both of which while not typically toxic, cause a variety of harmful effects.

Noise pollutions



Noise is unwanted sound. The main sources of noise are various modes of transportation (like air, road, rail-transportation), industrial operations, construction activities and celebrations (social/religious functions, elections etc) and also use of electric home appliances. Noise causes the following effects:

- interferes with man's communication - in a noisy area, communication is severely affected
- hearing damage - noise can cause temporary or permanent hearing loss. It depends on intensity and duration of sound level. Auditory sensitivity is reduced with noise level of over 90 dB in themed high frequency for more than a few minutes
- physiological and psychological changes – continuous exposure to noise affects the functioning of various systems of the body. It may result in hypertension, insomnia (sleeplessness), gastro-intestinal and digestive disorders, peptic ulcers, blood pressure changes, behavioural changes, emotional changes, etc.

Soil pollutions



Industrial wastes are the effluents discharged from chemical industries, paper and pulp mills, tanneries, textile mills, steel industries, distilleries, refineries, pesticides and fertilizer industries, pharmaceutical industries, food processing industries, cement industries, thermal and nuclear power plants, mining industries, etc. Thermal power plants generate a large quantity of, 'fly ash'. Huge quantities of these wastes are dumped on soils, thus

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contaminating them. Pesticides are used to kill pests that damage crops. These pesticides ultimately reach the soil and persist there for a long time. Pesticides which are persistent in nature are chlorinated hydrocarbon insecticides. Residues of these pesticides in the soils have long term effects especially under the temperate conditions. Industrial wastes also contain some organic and inorganic compounds that are refractory and non-biodegradable.

Industrial sludge may contain various salts, toxic substances, metals like mercury, lead, cadmium, arsenic etc. Agrochemicals released with the wastes of pesticide and fertilizer factories or during agricultural practices also reach the soil and pollute it. Soil also receives excreta from animals and humans. The sewage sludge contains many pathogenic organisms, bacteria, viruses and intestinal worms which cause pollution in the soil. The sources of radioactive substances in soil are explosion of radioactive devices, radioactive wastes discharged from industries and laboratories, aerial fall out, etc. Isotopes of radium, uranium, thorium, strontium, iodine, caesium and of many other elements reach the soil and persist there for a long time and keep on emitting radiations.

Some awareness activities and education that can be implemented at workplace that can contribute in conserving the environment are:

- Plant as many trees and plants as your space permits in the workplace vicinity - plants and trees absorb carbon dioxide and convert it into oxygen, thus helping to maintain carbon dioxide levels, but with deforestation taking place, carbon dioxide levels are only increasing. One tree can absorb approximately one tonne of carbon dioxide in its lifetime. Other than that, trees and plant helps as noise buffer.
- Implement waste management - waste management policies, which reduce greenhouse gas emissions include landfill gas capture and use and 'Refuse, Reuse, Recycle and Reduce' (4R) concept. Reducing refused production or using alternative means of disposal that reduce greenhouse gases emissions from waste decomposition and processing can have a large impact.
- Promote the usage of public transport than private vehicles - this can reduce number of vehicles on the road at one time, thus carbon emissions and also noise pollutions can be reduced. Vehicular pollution can be checked by regular tune-up of engines; replacement of more polluting old vehicles; installing catalytic converters; engine modification to have fuel efficient (lean) mixtures to reduce CO and hydro carbon emissions; and slow and cooler burning of fuels to reduce No_x emission

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- Encourage workers to use bicycle inside workplace vicinity.
- Reduce and conserve heat - heat loss is one of the biggest energy wastes. This can be reduced by lessen the use of glass on buildings, replacing all single glazed windows with double glazing or keeping all windows and doors closed during winter.
- Change lighting system - replacing light bulbs with compact fluorescent light bulbs that use 60% less energy.
- Conserve energy -switch off fans and lights when not used. Large amount of fossil fuels are burnt to produce electricity. Environment can be saved from degradation by reducing the amount of fossil fuels to be burned.
- Substitution of raw materials - by another purer grade raw material, which reduces the formation of pollutants.
 - (a) Low sulphur fuel, which has less pollution potential than high sulphur fuels
 - (b) Refined liquid petroleum gas (LPG) or liquefied natural gas (LNG) than traditional high contaminant fuels such as coal.

Awareness programs (National Level)

National Environmental Week, Malaysia

This event is celebrated from 21th to 27th October every year, as one of the most important activities by the Department of Environment(DOE) in promoting the environmental awareness at all levels of society in Malaysia. The main objective of the celebration is to:

- raise the level of environmental awareness among the public;
- strengthened and enhanced cooperation between the federal government and state governments in environmental management; and
- enhance cooperation between the various private agencies, non-governmental organizations (NGOs), individuals and others in environmental conservation.

Various activities undertaken to promote awareness of the industries and public to be more caring, sensitive and together play a role in protecting and conserving the environment.

Langkawi Award

Langkawi Award was introduced by the Malaysian Government to recognize and honor the contributions of organizations and citizens who have made a significant contribution to the environment and excel, whether in management, planning, trade, research, academic,

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publishing or fields related to the environment. The selection criteria is based on its:

- contribution to the promotion and participation of the community to protect the environment;
- solve specific environmental problems that seriously impact both locally and abroad; and
- mobilize action to solve environmental problems.

Langkawi Award was first awarded in 1991 and since then, except in 1996, several Malaysian organizations and individuals who excelled in their contribution to the environment has been awarded. Langkawi Award winner received a certificate of appreciation signed by the King, a plaque and a sum of RM 10,000.00.

Bandar Lestari –Environment Award

Department of Environment in collaboration with the Ministry of Housing and Local Government as well as several other government agencies have established Bandar Lestari - Environment Award, which was launched by the Minister of Science and Environment in conjunction with World Environment Day on 5th June 2003. The objective of this award is to:

- appreciate the efforts and contribution of local authorities which have integrated the environmental dimension into their development policies and actions;
- foster environmental awareness with the support of local authorities; and
- promoting innovative approaches and actions to achieve sustainable development (sustainable).

The selection criteria for winning this award is based on:

- physical environment (physical environment) - the level it takes to repair the physical environment such as improved water and air quality and reducing noise problems;
- initiative of ecology (ecological initiatives) - the element of the conservation of the natural environment such as habitat improvement and enhancement of biological diversity;
- municipal services (urban services) - Administration based on the environment (environmental governance) - the application of policies and procedures that promote transparent administration, two-way communication with the public and training an effective environmental management; and
- education and awareness - implement education and communication activities that

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enhance public awareness.

Sekolah Lestari - Environment Award

Sekolah Lestari - Environment Award is a joint program of Department of Environment, Ministry of Natural Resources and Environment, Ministry of Education and the Institute for Environment and Development (consolidated), Universiti Kebangsaan Malaysia (UKM). The program was officially launched by YB Dato' Sri Hj. Adenan Satem, Minister of Natural Resources and Environment and the Honourable Dato'Sri Najib Tun Razak, Minister of Education on 27th January, 2005. In conjunction with the event, the two documents entitled "Basic Formation Sekolah Lestari - Environment Award" and "Guidelines on Implementation and Evaluation of Sekolah Lestari - Environment Award" was launched to help the implementation and evaluation of this program in schools.

Sekolah Lestari is aims to establish a school environment that emphasizes an integrated approach to environmental protection and conservation in school management, curriculum, co-curriculum and greening. The involvement of the school work force in Sekolah Lestari activities give them the opportunity to practice, appreciate and cultivate eco-friendly features in the school environment, home, community, society and country. Sekolah Lestari is designed to achieve the following objectives:

- fostering environmental values amongst the school community;
- raise awareness amongst schools of the importance of preservation and conservation of the environment;
- encourage the school to implement environmentally friendly activities effectively;
- promote cooperation between schools and the community in the success of environmental education; and
- create a school environment that is conducive to producing a school that practices a sustainable way of life.

The award-winning Sekolah Lestari - Environment Award will receive a trophy and cash award ore RM10,000.00.

Projek Wira Alam

This is a joint project between the Department of Environment with the Ministry of Education and the Malaysian Nature Society (MNS). It was launched on 5th June 1998. Objectives of the programe:

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- provide opportunities for students to participate either individually or collectively in environmental activities;
- to inculcate students to learn about the environment;
- strengthen the students to communities in line with the concept of a caring society;
- to produce young people who have the expertise to take effective action, responsible and skilled to protect the environment as pemimo in future;
- contributions to protecting and caring for the environment.

In this program, each student have to complete the tasks defined in the Wira Alam activity book at every level before they can advancing to the next level, Level 1 (Wira Diri), Level 2 (Wira Komuniti) and Level 3 (Wira Alam).

Environmental Debate

The Department of Environment has established cooperation with the Majlis Debat Universiti-universiti Malaysia to organize the Environmental Debate between public higher education institutions in Malaysia. This program has been held since 1991. The objective of this debate is to:

- increase the awareness and knowledge of environmental issues among students so that they will be more sensitive to environmental problems;
- to train students as future leaders to think and express them selves effectively and thoughtful; and
- seek debate talent among students.

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SOALAN/QUESTION :

1. What are texample of environmental problems that can be happened nowadays due to uncontrolled activities by human?
2. In the workplace, what can the emploters should do to protect the environment?
3. What is environmetal education?
4. Name 5 types of environmental problems that mainly can be contributed by the industries?
5. Name 6 national environmental awareness program that have been conducted by the government in supporting the conservation of environment?
6. Explain 5 objectives of the Sekolah Lestari – Environment Awareness conducted at the national level for the school.

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